Type: DSC2-5 (Minor) B.Sc(ECS)-II (Semester IV) Course Title: Statistics for Data Science-II (Paper Code:)		
Credits: Theory – (2)		**Practical's – (1)**
Total Lectures: 30 Hrs.		**Contact Hrs. (L) :** 2
University Evaluation: 30 Marks		**Internal Evaluation:** 20 Marks
Course Outcomes:		
1.	To understand the basic probability concepts and random variables.	
2.	To understand the concept of a random sample from a distribution, sampling distribution of a statistic, standard error of important estimates such as mean and proportions.	
3.	To understand various basic concepts on sampling distributions and large sample tests based on normal distribution.	
4.	To get knowledge about inferences from Binomial and Poisson distributions as illustrations.	
5.	To get knowledge about important inferential aspects such as point estimation, test of hypotheses and associated concepts.	
6.	To understand the chi-square test for independence of attributes and goodness of fit.	
Unit	**Content**	**Lect.**
I	**Permutations and Combinations:** Principles of counting, Permutations of 'n' dissimilar objects taken 'r' at a time (with and without replacement), Permutations of 'n' objects not all different, Combinations of 'r' objects taken from 'n' objects, Numerical problems. **Probability:** Idea of deterministic and non-deterministic models, Sample space: types of sample space, Event: types of events, Classical definition of probability and its limitations, Axiomatic definition of probability, numerical examples. **Proofs of results:** $P(\phi) = 0$, $P(A^c) = 1-P(A)$, $P(AUB) = P(A) + P(B) - P(A \cap B)$, $P(A) \leq P(B)$ if A is subset of B. **Conditional probability:** concept and definition, multiplication law of probability. **Independence of events:** concept and definition, pairwise and complete independence (for 3 events), Partition of sample space and Baye's theorem, Numerical problems.	**10**
II	**Discrete Random Variable and standard Discrete Distributions:** Definition of discrete random variable(r.v.), Definition of Probability mass function(p.m.f.) , cumulative distribution function(c.d.f.), properties of c.d.f **Mathematical Expectation and variance of r.v.:** definition, Numerical problems.	**20**

	Standard Discrete Distribution: a) Binomial Distribution: Definition, mean and variance (Statement only), real-life situations, adaptive property and numerical examples. b) Poisson Distribution: Definition, mean and variance (Statement only), real life situations, adaptive property and numerical examples. **Contentions Random variable and Standard Contentions Distributions:** Definition of Contentions random variable(r.v.), Definition of Probability density function(p.d.f.) , cumulative distribution function(c.d.f.), properties of c.d.f **Mathematical Expectation and variance of r.v.:** definition, Numerical problems. **Standard Contentious Distributions:** a) **Uniform distribution:** definition, mean and variance, c.d.f., probability curve and numerical examples. **b) Normal distribution:** definition, mean and variance, probability curve, Standard Normal Variable(S.N.V.), properties of normal distribution, distribution of (aX + b), (aX + bY + c) when X and Y are independent variables, computations of probabilities using normal tables. **c) Sampling Distributions and Test of Hypotheses:** Concepts of parameter, statistic, sampling distribution of a statistic, standard error. Sampling distribution of sample mean, standard errors of the sample mean, sample variance, and sample proportion. Null and alternative hypotheses, level of significance, Type I and Type II errors, their probabilities and critical region, confidence intervals, and p value. Tests of significance and confidence intervals for - single proportion, single mean. **Statistical Inference:** Estimation and Hypothesis Testing (only concept). **Hypothesis Testing for a Single Population:** The concept of hypothesis testing tests involving a population mean and population proportion (z test and t test). Chi square test for independence of attributes and goodness of fit.	
Reference Books:		
1.	Fundamentals of Statistics by Goon Gupta, Das Gupta.	
2.	Statistical Methods by S. P. Gupta	
3.	Business Statistics by S. Shaha	
4.	Fundamentals of Mathematical Statistics by Kapoor and Gupta	
5.	Programmed Statistics by B. L. Agarwal	
6.	Statistical Methods by P. N. Arora,Summeet Arora, S. Arora	
7.	Introduction to discrete probability and probability distributions by Madhav B. kulkarni, Surendra B. Ghatpande.	